Autoliv AUTOLIV (CHINA) ELECTRONICS	Brake Fluid Evacuation and Fill Process Specification	Date: 2014.10.08 Rev.: D

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Brake Fluid Evacuation and Fill Process Specification
制动液抽真空加注工艺规范

SUBJECT: This document defines the process flow for brake fluid evacuation and fill and all related diagnostic communication services.
这份文件定义了制动液抽真空加注工艺流程以及相关联通讯信号的详细信息。

REVISION HISTORY 版本历史

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A	Kailai Sun	2014.08.26	Initial Release
B	Kailai Sun	2014.09.11	Delete security access service, update some description
C	Kailai Sun	2014.09.30	Update 2F service content
D	Kailai Sun	2014.10.08	Update 2F service request and response format

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Reference Documents 参考文档

The following is a list of documents used to establish this document.

Nº	Document	Reference	Source
1			
2			

1 Terminology 专用术语

The following is a list of terms and acronyms that are in the document.
以下为本文档涉及的相关专用术语

IRBC – Integrated Restraints and Brake Controller 约束和制动整合控制器

SHA - Smart Hydraulic Actuator 智能液压控制器

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2 Introduction 介绍

Brake fluid evacuation and fill is a crucial process to ensure the brake performance during vehicle operation. Brake lines need to be evacuated before filling with brake fluid. This keeps out air trapped in brake lines after brake fluid filling. In addition, this also prevents the brake fluid contamination from moisture and dissolved air. This document defines the process flow for brake fluid evacuation and fill, as well as all related diagnostic communication services.

制动液抽真空加注是一道确保车辆制动性能的关键工艺。制动油管需要在加注制动液前被抽真空以确保加注完后不会有空气残留在油管中。这样也可以有效控制制动液在加注过程中的空气溶解度以及防止其他污染物的导入。本文档定义了制动液抽真空加注的工艺流程以及相关的通讯信号。

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3 Brake Fluid Requirement 制动液型号要求

DOT3 or DOT4 brake fluid shall be used in the evacuation and fill process to ensure the brake system performance.

为确保制动系统的工作性能，抽真空加注中制动液必须使用 DOT3 或者 DOT4 型号。

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4 Evacuation and Fill Process Procedure 抽真空加注工艺流程

The evacuation and fill process consists of five steps. Vacuum level and brake fluid pressure in the brake system are measured as control parameters to monitor the process. Communication messages are sent to IRBC to control the On/Off of prime valves, release valves, and SHA pump motor throughout the entire process. A graphic view on the process is provided in Figure A in the Appendix. Section 6 explains more details on all the communication messages related in the evacuation and fill process.

抽真空加注流程一共包含有五个步骤。加注设备通过测量制动系统中的真空度和制动液压力这两个参数来监控整个抽真空价值过程。

SHA 内部的供油阀、降压阀和电机需要在制动液抽真空加注过程中适时开关，这是通过向 IRBC 发送相应的通讯信号来实现的。附件中的图 A 通过图表的方式描述了加注工艺中的各项参数要求。本文件第 6 节对所有涉及到的通讯信号做了详细定义。

4.1 Air Pressure Decay Test (Step 1) 气密性检测

The air pressure decay test is to evaluate air tightness of the brake system. Air is pumped into the master cylinder reservoir and air pressure in the brake system is measured to validate the air tightness of the brake system.

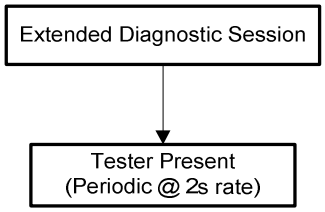
气密性检测的目的是确保整个制动系统管路的密封性。通过向主缸储油罐注入空气，检测气压的变化来检验制动系统的密封性。

4.2 Pre-Evacuation (Step 2) 前抽真空

Before evacuating the brake system, the vehicle needs to be powered on and establish communication between the IRBC and process equipment through OBDII connector. The process equipment sends out 2 messages in sequence (as defined in Figure 1) to IRBC.

在开始抽真空前，车辆需要上电源并且确保加注设备通过 OBDII 接口与车辆建立通讯。加注设备会向 IRBC 依次发送两个报文（见图 1）。

Figure 1



4.3 Evacuation (Step 3) 抽真空

The process equipment starts to evacuate the brake system. Vacuum level measured at test head shall be maintained at 0.67-0.13 mbar. Once the vacuum level in the brake system stabilizes at < 5 mbar, the process equipment send messages to the IRBC that results in

加注设备开始对制动系统进行抽真空。在设备的抽真空探头处测量的真空度应保持在 0.67-0.13 mbar。当制动系统内的真空度稳定在< 5 mbar 时，加注设备向 IRBC 发送报文按照附录中图 A 描述的方式按照 1 秒的间隙来

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alternately turning on and off SHA release valves, prime valves, and pump motor with 1s interval as described in Figure A in Appendix. Actual time to activate valves and pump motor during the evacuation varies depending on the equipment and vehicle type. Discussion is required to properly define this timing.

交替开启和关闭 SHA 所有的供油阀、降压阀和电机。抽真空过程中，具体执行此电磁阀和电机开启/关闭的时机因设备和车型而异，需进行商讨决定合理的时机。

4.4 Brake Fluid Fill (Step 4) 制动液加注

Brake fluid is filled into brake system with minimum fill pressure > 4 bar, while the maximum fill pressure is limited by the pressure capability of the master cylinder reservoir. SHA release valves, prime valves, and pump motor shall continue stay in alternatively on and off with 1 s interval status. Duration for valves and pump motor alternate on and off shall not exceed 60s with vehicle battery at 12.5V through the entire evacuation and fill. Actual time to turn off valves and pump motor during the fill varies depending on the equipment and vehicle type. Discussion is required to properly define this timing.

制动液通过加注设备注入制动系统中。加注的压力应> 4 bar，最大的加注压力取决于主缸储油罐的最大承压能力。SHA 的供油阀、降压阀和电机应当继续保持 1 秒的间隙交替开启和关闭状态。在整个抽真空加注过程中，交替执行开启和关闭 SHA 所有的电磁阀和电机的总时间在车辆电瓶电压为 12.5V 的情况下，不能够超过 60 秒。具体执行此电磁阀和电机关闭的时机因设备和车型而异，需进行商讨决定合理的时机。

4.5 Completion (Step 5) 完成

The tester records a pass/fail result for the evacuation and fill process. If result is Pass, the process equipment sends the Evacuation and Fill Complete DID message. Then the process equipment sends Default Session message to IRBC and vehicle power is turned off.

加注设备记录最后抽真空加注的成功/失败结果。如果加注工序成功完成，加注设备向 IRBC 发送报文记录抽真空加注成功的 DID 在 IRBC 内并重置 IRBC，然后发送 Default Session 报文。最后，断开车辆电源，抽真空加注工序结束。

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5 Communication 通讯信号

All CAN communication shall use CAN ID 0x731 to send messages to IRBC and CAN ID 0x739 to receive responses.

所有的 CAN 通讯应使用 CAN ID 0x731 来向 IRBC 发送报文，然后使用 CAN ID 0x739 来接收来自 IRBC 的反馈。

5.1 Diagnostic Service

The diagnostic session control service enables diagnostic sessions in the IRBC. A diagnostic session defines applications required for the evacuation and fill process. Messages for diagnostic service are specified as follows:

诊断会话控制服务用于在 IRBC 中开启一个诊断会话。一个诊断会话定义了一系列抽真空加注过程所需要的应用。相关的报文信息定义如下：

5.1.1 Request Message Definition

Process Equipment to IRBC

Data Byte	Parameter Name	Message Usage	Hex Value
#1	Diagnostic Session Control Request Service ID	M	10
#2	Sub Function = [Diagnostic Session Type] Default Session Extended Diagnostic Session	M	 01 03

5.1.2 Positive Response Message

IRBC to Process Equipment

Data Byte	Parameter Name	Message Usage	Hex Value
#1	Diagnostic Session Control Response Service ID	M	50
#2	Sub Function = [Diagnostic Session Type] Default Session Extended Diagnostic Session	M	XX 01 03
 #3 #4 #5 #6	Session Parameter Record [] =[P2Can_ECU_max (high byte) P2Can_ECU_max (low byte) P2*Can_ECU_max (high byte) P2*Can_ECU_max (low byte)]	 M : : M	 00-FF : : 00-FF

5.1.2.1 Session Parameter Record Definition

Parameter	Definition	Min	Max
P2Can_ECU_max	Performance requirement for the ECU to start with the response message after the reception of a request message.	0 ms	50 ms
P2*Can_ECU_max	Performance requirement for the ECU to start with a further response message after the transmission of a negative response message.	0 ms	5000 ms

5.1.3 Negative Response Message

IRBC to Process Equipment

Data Byte	Parameter Name	Message Usage	Hex Value
#1	Negative Response	M	7F

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#2	Diagnostic Session Control Request Service ID	M	10
#3	Response Code = [Sub-Function Not Supported Incorrect Message Length Or Invalid Format Condition Not Correct]	M	 12 13 22

5.2 Tester Present

This service provides indication to IRBC that a diagnostic test tool is connected to the CAN bus and diagnostic sessions and communication have been activated. Tester Present message needs to be sent at a two second periotic rate starting at Step 2 of the evacuation and fill process and ending at Step 8.

此项服务用来提示 IRBC 诊断工具已经连接上 CAN 总线，诊断对话和通讯处于激活状态。Tester Present 报文需要在抽真空加注过程的步骤 2 开始以 2 秒发送一次的周期发送给 IRBC，一直到步骤 8 结束为止。

5.2.1 Request Message Definition

Process Equipment to IRBC

Data Byte	Parameter Name	Message Usage	Hex Value
#1	Tester Present Request Service Id	M	3E
#2	Sub-Function = [Zero Sub-Function] Positive Response Required No Positive Response Required	M	 00 80

5.2.2 Positive Response Message

IRBC to Process Equipment

Data Byte	Parameter Name	Message Usage	Hex Value
#1	Tester Present Response Service ID	M	7E
#2	Sub-Function = [Zero Sub-Function] Positive Response Required		 00

5.2.3 Negative Response Message

IRBC to Process Equipment

Data Byte	Parameter Name	Message Usage	Hex Value
#1	Negative Response	M	7F
#2	Tester Present	M	3E
#3	Response Code = [Sub-Function Not Supported Incorrect Message Length Or Invalid Format]	M	 12 13

5.3 Input Output Control by Identifier

The InputOutputControlByIdentifier service is primarily used by the IRBC to control the SHA in the brake system. The process unit will send IRBC Request message, which contains a dataIdentifier to reference a group of signals of the IRBC. The IRBC shall send a Positive

IRBC 通过 InputOutputControlByIdentifier 服务来控制 SHA 内部的阀门和电机。加注设备向 IRBC 发送 Request 报文，此请求报文中包含有一个引用了一系列 IRBC 用于控制 SHA 信号的 dataIdentifier。如果请求报文成功被执行

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Response Message if the Request Message can be successfully executed.

The SHA enabled by Input and Output Control commands will remain active for a while (settling time) even after it has been de-activated by Input and Output Control command. When any of the following conditions are met, the IRBC will begin to terminate this service.

- Positively responded to a new diagnostic session request.
- Any Output Control limit is exceeded. See section 6.4.3 Negative Response Message Data Bytes
- The active session is timed out after 5s without receiving any request message (include Tester Present message).

The IRBC will remain in the Input and Output Control until it is determined that the commanded SHA is no longer active. The IRBC will then return control to normal operation. During the SHA settling time, the IRBC is allowed to go back to this service.

了，IRBC 会向加注设备发送一个 Positive Response 报文。

即便 Input and Output Control 已经解除了 SHA 的激活状态，先前被此指令启动的 SHA 会依旧保持一段时间（稳定时间）的激活状态。当满足以下任意一个条件时，IRBC 会终止 InputOutputControlByIdentifier 服务：

- IRBC 正确响应了一个新的会话诊断请求
- 超出任意第 6.4.3 节中定义 Output Control 极值
- 激活的对话没有接收到任何诊断请求报文(包括 Tester Present)超过 5 秒钟

IRBC 会处在 Input and Output Control 服务中，一直到 SHA 不再处于激活的状态。然后 IRBC 会返回到正常工作状态。当还 SHA 处在稳定时间时，IRBC 还可以随时返回到此服务中。

5.3.1 Request Message Definition

Process Equipment to IRBC (shortTermAdjustment)

Data Byte	Parameter Name	Cvt	Hex Value
#1	inputInputOutputControlByIdentifier Request Sid	M	2F
#2 #3	dataIdentifier #1 [] = [byte #1 (MSB) byte #2 (LSB)]	M	 31 07
#4	inputOutputControlParameter=[shortTermAdjustment]	M	 03
 #5 #6 #7	ControlOptionRecord #1 = [ControlState#1 ControlState#2 ControlState#3]	 M M M	 00-FF 00-FF 00-FF

Process Equipment to IRBC (returnControlToECU)

Data Byte	Parameter Name	Cvt	Hex Value
#1	inputInputOutputControlByIdentifier Request Sid	M	2F
#2 #3	dataIdentifier #1 [] = [byte #1 (MSB) byte #2 (LSB)]	M	 31 07

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#4	inputOutputControlParameter=[returnControlToECU]	M	 00

5.3.1.1 Control State Definition

Control State	Bit #	Definition	Units	Range	Command Resolution
#1	7	Rear Right Release Solenoid On-Off Control Command	N/A	N/A	On = 1, Off=0
	6	Rear Right Apply Solenoid On-Off Control Command	N/A	N/A	On = 1, Off=0
	5	Rear Left Release Solenoid On-Off Control Command	N/A	N/A	On = 1, Off=0
	4	Rear Left Apply Solenoid On-Off Control Command	N/A	N/A	On = 1, Off=0
	3	Front Right Release Solenoid On-Off Control Command	N/A	N/A	On = 1, Off=0
	2	Front Right Apply Solenoid On-Off Control Command	N/A	N/A	On = 1, Off=0
	1	Front Left Release Solenoid On-Off Control Command	N/A	N/A	On = 1, Off=0
	0	Front Left Apply Solenoid On-Off Control Command	N/A	N/A	On = 1, Off=0
#2	7	Primary Prime Solenoid On-Off Control Command	N/A	N/A	On = 1, Off=0
	6	Secondary Prime Solenoid On-Off Control Command	N/A	N/A	On = 1, Off=0
	5	Primary Isolation Solenoid On-Off Control Command	N/A	N/A	On = 1, Off=0
	4	Secondary Isolation Solenoid On-Off Control Command	N/A	N/A	On = 1, Off=0
	3	Not Used.	N/A	N/A	N/A
	2	Not Used.	N/A	N/A	N/A
	1	Pump Motor On/Off Control Command	N/A	N/A	On = 1, Off=0
	0	System Relay Control Command	N/A	N/A	On = 1, Off=0
#3	All	Not Used.	N/A	N/A	N/A

5.3.2 Positive Response Message

IRBC to Process Equipment (shortTermAdjustment)

Data Byte	Parameter Name	Cvt	Hex Value
#1	inputInputOutputControlByIdentifier Positive Response Sid	S	6F
 #2 #3	dataIdentifier #1 [] = [byte #1 (MSB) byte #2 (LSB)]	M	 31 07
#4	inputOutputControlParameter=[shortTermAdjustment]	M	 03
 #5 #6 #7	ControlOptionRecord #1 = [ControlState#1 ControlState#2 ControlState#3]	 M M M	 00-FF 00-FF 00-FF

IRBC to Process Equipment (returnControlToECU)

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Data Byte	Parameter Name	Cvt	Hex Value
#1	inputInputOutputControlByIdentifier Positive Response Sid	S	6F
#2 #3	dataIdentifier #1 [] = [byte #1 (MSB) byte #2 (LSB)]	M	31 07
#4	inputOutputControlParameter=[returnControlToECU]	M	00

5.3.3 Negative Response Message

IRBC to Process Equipment

Data Byte	Parameter Name	Cvt	Hex Value
#1	negativeResponse Service Id	S	7F
#2	inputInputOutputControlByIdentifier Request Sid	M	2F
#3	Response Code = [Incorrect Message Length Or Invalid Format Conditions Not Correct Request Out Of Range Service Not Supported In Active Session Motor Or Solenoid Thermal Limit Exceeded Vehicle Speed Limit Exceeded System Voltage Out Of Range]	M	 13 22 31 7F 86 88 93

5.3.3.1 Response Code Definition

Hex	Condition
13	**incorrectMessageLengthOrInvalidFormat** Wrong length for the specific request
22	**ConditionsNotCorrect** This response code shall be sent if the operating conditions of the ECU are such that it can't perform the required action. The system is currently in initialization or shutdown process. The system is in the process of changing memory.
31	**RequestOutOfRange** Wrong inputOutputLocalIdentifier
7F	**serviceNotSupportedInActiveSession** The service is not enabled in the current diagnostic session.
86	**temperatureTooHigh** This response code indicates that the requested action will not be taken because the ECU prerequisite condition for temperature is not met (current temperature is above a pre-programmed maximum threshold).
88	**vehicleSpeedTooHigh** This response code indicates that the requested action will not be taken because the ECU prerequisite condition for vehicle speed is not met (current vehicle speed is above a pre-programmed maximum threshold).
93	**voltageTooLow** This response code indicates that the requested action will not be taken because the ECU prerequisite condition for voltage at the primary pin of the ECU (ECU) is not met (current voltage is below a preprogrammed maximum threshold).

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5.4 Examples CAN Messages for SHA Control

The process equipment sends CAN messages to IRBC to control the On/Off of release valve, prime valve and pump motor of the SHA. These CAN messages are defined as follows:

加注设备通过向 IRBC 发送 CAN 报文来控制 SAH 的供油阀、降压阀和电机的开关。CAN 信号报文定义如下：

5.4.1 Release Valve and Prime Valve On and Motor On

Process Equipment to IRBC (Periodic at 100ms rate)

ID	DLC	Data							
		byte 0	byte 1	byte 2	byte 3	byte 4	byte 5	byte 6	byte 7
0x731	8	0x07	0x2F	0x31	0x07	0x03	0xAA	0xC3	0x00

5.4.2 Return to Normal State

Process Equipment to IRBC (Periodic at 100ms rate)

ID	DLC	Data							
		byte 0	byte 1	byte 2	byte 3	byte 4	byte 5	byte 6	byte 7
0x731	8	0x04	0x2F	0x31	0x07	0x00	0x00	0x00	0x00

5.5 Evacuation and Fill Complete DID

This service is used to write DID in the IRBC to indicate that the evacuation and fill process is complete and passed.

此项服务用于待抽真空加注完成后在 IRBC 的 DID 中写入数值以表示加注已完成。

5.5.1 Request Message Definition

Process Equipment to IRBC

Data Byte	Parameter Name	Message Usage	Hex Value
#1	Write DID Request Service ID	M	2E
#2 - #3	DID Location	M	2D04
#4	DID Value	M	AA

Note: 0x00: End-of-line test not completed
0xAA: End-of-line test completed and ok
0xEE: End-of-line test completed and not ok

5.5.2 Positive Response Message

IRBC to Process Equipment

Data Byte	Parameter Name	Message Usage	Hex Value
#1	Write DID Response Service ID	M	6E
#2 - #3	DID Location	M	2D04

5.5.3 Negative Response Message

IRBC to Process Equipment

Data Byte	Parameter Name	Message Usage	Hex Value

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#1	Negative Response	M	7F
#2	Tester Present	M	2E
#3	Response Code = [Incorrect Message Length Or Invalid Format Condition Not Correct Request out of Range Security Access Denied]	M	 13 22 31 33

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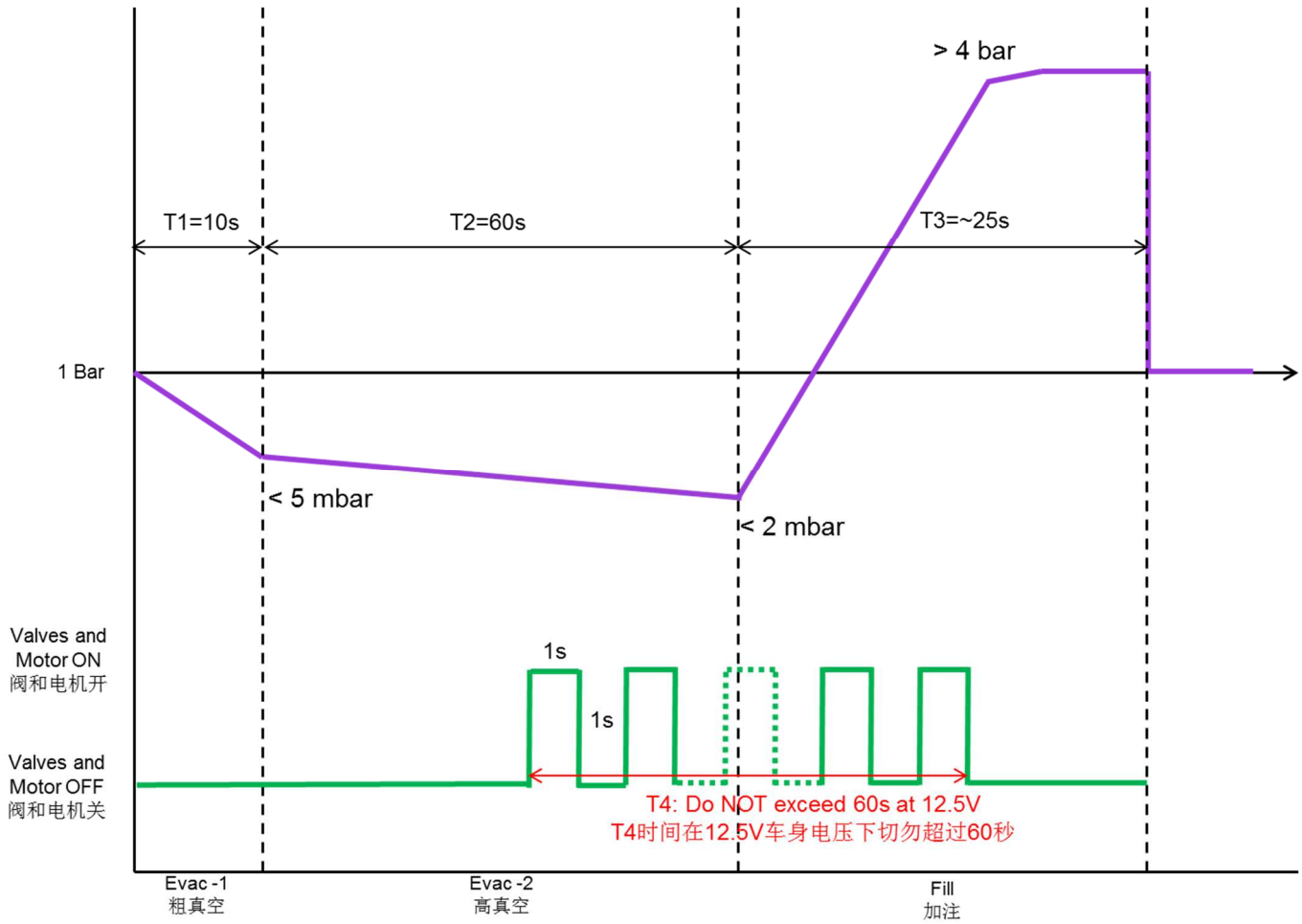
6 Appendix 附件

Figure A: Brake Fluid Evacuation and Fill Process Graph



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